Satellite Networks under Typed Frames and Temporal Observation Channels

Rigid-frame covariance, torus closures, line-of-sight graphs, sampling, and clock protocols

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Status. This document supersedes the bilingual working note v1.1 and is an application-layer reference migrated to GO Core v0.2. It is not an orbit-determination method, an operational constellation model, a replacement for TLE/SGP4, or a new result in astrodynamics. Its contribution is a typed separation of hidden state, coordinate frame, sensor channel, temporal graph, sampling protocol, closure, and relativistic clock comparison.

Abstract

A satellite constellation is represented by one labeled dynamical state but by many coordinate traces and still more sensor records. The legacy note treated all of these as observer-frame drawings and then called pairwise distances, coordinate spectra, orbit closures, density statistics, and clock offsets "invariants". This reference states the transformation group and protocol for every such claim. For a time-dependent rigid frame $\mathfrak{f} = (o, R) \in C^1(I, \mathrm{SE}(3))$, the labeled Euclidean distance matrix is invariant at each common time, while component velocities, coordinate spectra, and single-object orbit closures generally are not. A distance matrix determines a labeled configuration only up to an element of $E(3)$, so chirality is lost. For a quasiperiodic phase flow, the exact phase closure is the coset subtorus determined by the integer resonance lattice; its image under a continuous observation map is channel-dependent and is only equivariant under a constant rigid isometry. Finite sampling introduces exact frequency aliases and cannot certify rational independence. Spherical-body line-of-sight and range gates define a temporal graph whose snapshot and time-respecting connectivity are separated. Proper time along a fixed worldline segment is a spacetime scalar, whereas comparing different clocks at equal coordinate time requires a declared synchronization and gravity model. These distinctions close the last frame-law defect in the Geometry of Observation corpus.

Contents

1 Scope, hidden state, and the observation chain

Fix a coordinate-time interval $I \subset \mathbb{R}$, a finite label set $V = \{1, \ldots, N\}$, and an Earth-centered inertial reference chart. The minimal spatial state is the labeled tuple

$$X(t) = \big(c_E(t), (r_i(t), v_i(t))_{i \in V}\big), \qquad v_i = \dot{r}_i, \tag{1}$$

where c_E is the center of the occulting body. A set notation $\{r_i\}$ is insufficient: it discards labels and multiplicity, both of which are required for tracking and temporal edges.

Model 1.1 (Two-body benchmark)**.** The reproducibility controls use

$$\dot{r}_i = v_i, \qquad \dot{v}_i = -\mu_E \frac{r_i - c_E}{\|r_i - c_E\|^3}, \tag{2}$$

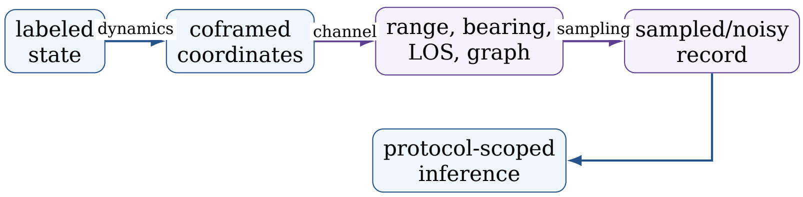
with a spherical exclusion radius R_E. This is an exact model definition, not an ephemeris accuracy claim.

Classical elements $(a, e, i, \Omega, \omega, M_0)$ are a local coordinate chart for model 1.1; they are singular at standard circular and equatorial limits and are not the hidden object. Likewise, a two-line element set is a model-specific mean-element record intended for a compatible SGP4-family propagator, not an arbitrary Kepler state. The former phrase "TLE-like benchmark" is therefore removed.

Definition 1.2 (Typed observation protocol)**.** A protocol is the composite

$$\mathcal{X} \xrightarrow{\Phi_t} \mathcal{X}_t \xrightarrow{F_\mathfrak{f}} \mathcal{Y}_\mathfrak{f} \xrightarrow{\mathcal{Q}} \mathcal{D} \xrightarrow{\mathcal{S}_{\Delta t,T}} \mathcal{D}^M \xrightarrow{\mathcal{A}} \mathcal{I}, \tag{3}$$

where Φ_t is the declared dynamics, $F_\mathfrak{f}$ is an invertible coordinate change, $\mathcal{Q}$ is a possibly lossy sensor or graph channel, $\mathcal{S}_{\Delta t,T}$ samples over a finite horizon, and $\mathcal{A}$ is an estimator or diagnostic. Information loss is attributed to $\mathcal{Q}$ or $\mathcal{S}$, not to a frame change.



2 Time-dependent rigid frames and exact spatial invariants

Let $o : I \to \mathbb{R}^3$ be a frame origin and let $R : I \to \mathrm{SO}(3)$ map frame components to inertial components. The coordinates of satellite i and of the Earth center are

$$y_i = R^\mathsf{T}(r_i - o), \qquad c_\mathfrak{f} = R^\mathsf{T}(c_E - o). \tag{4}$$

Thus a frame is not merely an observer point. Position vectors may be composed across an Earth–Sun–barycentric tree only after all origins, orientations, epochs, and time scales are declared. In homogeneous coordinates the legal composition is matrix multiplication,

$$T_{A \leftarrow C}(t) = T_{A \leftarrow B}(t) T_{B \leftarrow C}(t), \tag{5}$$

not an untyped sum of vectors stored in different bases.

Let

$$\mathcal{G}_I = C^1(I, \mathrm{SE}(3))$$

act diagonally on every node and on the central body through equation (4).

Theorem 2.1 (Distance-matrix invariance)**.** *For every $\mathfrak{f} \in \mathcal{G}_I$, every common time t, and every $i, j \in V$,*

$$D_{ij}^{\mathfrak{f}}(t) = \|y_i(t) - y_j(t)\| = \|r_i(t) - r_j(t)\| = D_{ij}(t). \tag{6}$$

Consequently, every deterministic function of the labeled distance matrix is invariant under this declared action.

Proof. Translations cancel and $R^{\mathsf{T}} R = I$:

$$y_i - y_j = R^{\mathsf{T}}(r_i - r_j), \qquad \left\| R^{\mathsf{T}}(r_i - r_j) \right\|^2 = (r_i - r_j)^{\mathsf{T}} R R^{\mathsf{T}} (r_i - r_j).$$

□

The corresponding velocity law is different. With $\Omega^{\times} = R^{\mathsf{T}} \dot{R} \in \mathfrak{so}(3)$,

$$\dot{y}_i = R^{\mathsf{T}}(v_i - \dot{o}) - \Omega^{\times} y_i. \tag{7}$$

Therefore coordinate velocities, accelerations, and component spectra are not invariants of $\mathcal{G}_I$. Even the norm of a relative velocity acquires the rotational term unless R is constant.

Theorem 2.2 (What the distance matrix identifies)**.** *Let D be the labeled Euclidean distance matrix of N points, let $J = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^{\mathsf{T}}$, and define*

$$B = -\frac{1}{2} J (D \circ D) J. \tag{8}$$

Then B is the centered Gram matrix. If $B \succeq 0$ and $\operatorname{rank} B \le 3$, its factorization reconstructs the configuration up to $E(3)$. Pairwise distances do not distinguish translations, rotations, or reflections.

Proof. Centering the points gives $B = X_c X_c^{\mathsf{T}}$. Any two factorizations of the same positive semidefinite Gram matrix differ by an orthogonal map on the realized span. Restoring the centroid adds an arbitrary translation. □

Thus the distance channel is complete for a labeled snapshot modulo $E(3)$, but not modulo $\mathrm{SE}(3)$: chirality requires an oriented volume or another parity-sensitive channel.

3 Phase closure, image closure, and frame covariance

For a quasiperiodic idealization, let

$$\theta(t) = \theta_0 + \omega t \pmod{2\pi}, \qquad \theta \in \mathbb{T}^k, \tag{9}$$

and define the integer resonance lattice

$$L_\omega = \{m \in \mathbb{Z}^k : m \cdot \omega = 0\}. \tag{10}$$

Theorem 3.1 (Exact phase closure)**.** *The closure of the phase orbit is the coset subtorus*

$$\Gamma_{\theta_0,\omega} = \{\theta \in \mathbb{T}^k : m \cdot (\theta - \theta_0) = 0 \pmod{2\pi} \text{ for every } m \in L_\omega\}, \tag{11}$$

with
$$\dim \Gamma_{\theta_0,\omega} = k - \operatorname{rank}_{\mathbb{Z}} L_\omega.$$
For every continuous $F : \mathbb{T}^k \to \mathbb{R}^d$*,*
$$\mathrm{cl}\{F(\theta(t)) : t \in \mathbb{R}\} = F(\Gamma_{\theta_0,\omega}). \tag{12}$$

The rationally commensurate and rationally independent cases are only the two endpoints of theorem 3.1; intermediate resonance rank is possible. Compactness and continuity justify equation (12). A noncompact secular drift is not silently treated as another torus angle.

Proposition 3.2 (Closure is equivariant, not invariant)**.** *For a constant* $g = (a, R) \in \mathrm{SE}(3)$ *and* $F_g(\theta) = a + RF(\theta)$*,*
$$K_{F_g} = gK_F. \tag{13}$$
Equality $K_{F_g} = K_F$ *requires an additional symmetry of the set. For a time-dependent* $g(t)$*, even congruence may fail.*

Proof. A homeomorphism commutes with closure, giving the first statement. For the second, take $F(\theta(t)) \equiv 0$ and $a(t) = (\cos \alpha t, \sin \alpha t, 0)$. The original closure is one point and the transformed closure is a circle. □

The protocol-invariant object is therefore the phase-flow resonance lattice, when the dynamical model and time parameter are fixed. The single-object spatial closure K_F is a channel output.

4 Sampling, retarded observation, and non-identifiability

At times $t_m = t_0 + m\Delta t$, $m = 0, \ldots, M-1$, the record is finite. Its support
$$\widehat{K}_{M,\Delta t} = \{Q(X(t_m)) : 0 \le m < M\} \tag{14}$$
is not the infinite-time closure in equation (12).

Proposition 4.1 (Exact sampling alias)**.** *For any* $q \in \mathbb{Z}$ *and* $t_m = m\Delta t$*,*
$$e^{i(\omega + 2\pi q/\Delta t)t_m} = e^{i\omega t_m}. \tag{15}$$
Hence a sampled coordinate spectrum does not identify an unrestricted continuous frequency.

Finite noisy data also cannot decide whether a frequency ratio is exactly rational: arbitrarily close rational and irrational vectors produce arbitrarily close records on a fixed horizon. Statements about closure dimension therefore require either a declared exact model or uncertainty-bounded inference.

The distinction between state and sensor time is equally important. For a reception event $(t_r, o(t_r))$, a one-way ideal observation uses an emission time $t_{e,i}$ satisfying
$$t_r - t_{e,i} = \frac{\|r_i(t_{e,i}) - o(t_r)\|}{c}. \tag{16}$$
Different satellites are generally seen at different emission times. Pairwise distances between apparent retarded positions are not the simultaneous distance matrix in theorem 2.1.

Bearing-only data
$$b_i(t) = \frac{y_i(t)}{\|y_i(t)\|}$$

discard range. Even without noise, one viewpoint therefore does not recover $D(t)$ unless additional baselines, ranging, or dynamical hypotheses are supplied.

5 Line-of-sight and temporal network geometry

Let $p_i = r_i - c_E$, $d_{ij} = p_j - p_i$, and let the spherical occulting radius including a declared margin be $R_{\text{occ}} > 0$. For $i \neq j$, define

$$u_{ij}^* = \text{clip}\left(-\frac{p_i \cdot d_{ij}}{\|d_{ij}\|^2}, 0, 1\right), \tag{17}$$

$$h_{ij} = \left\|p_i + u_{ij}^* d_{ij}\right\|. \tag{18}$$

The degenerate case $p_i = p_j$ is handled separately. A strict spherical line of sight is declared when

$$\text{LOS}_{ij}(t) = \mathbf{1}\{h_{ij}(t) > R_{\text{occ}}\}. \tag{19}$$

Tangency belongs to the blocked class in this convention.

For link range $R_{\text{link}} > 0$, the symmetric snapshot graph has

$$A_{ij}(t) = \mathbf{1}\{i \neq j\}\mathbf{1}\{D_{ij}(t) \leq R_{\text{link}}\}\,\text{LOS}_{ij}(t). \tag{20}$$

Because the satellites, the Earth center, and the segment are all transformed together, equation (20) is invariant under $\mathcal{G}_I$. A fixed spatial bin or an untransformed occulting center would break that statement.

Under a vertex relabeling by a permutation matrix P,

$$A \longmapsto PAP^\mathsf{T}, \qquad L_G = \text{diag}(A\mathbf{1}) - A \longmapsto PL_GP^\mathsf{T}.$$

Connected-component counts and $\text{spec}(L_G)$ are label-invariant, while node-specific routing outputs are equivariant under the same relabeling.

Definition 5.1 (Time-respecting journey)**.** Let an active edge $e_k = (v_{k-1}, v_k)$ be entered at t_k and have a declared nonnegative transmission delay $\ell_{e_k}(t_k)$. A journey satisfies

$$t_{k+1} \geq t_k + \ell_{e_k}(t_k) \tag{21}$$

for every consecutive edge. Temporal reachability and earliest arrival are defined by such journeys, not by connectivity of the time-aggregated graph.

The range-only graph $\mathbf{1}\{D_{ij} \leq \rho\}$ is called a *proximity graph*. It is not a conjunction-risk calculation. Operational collision assessment additionally requires predicted relative states, covariances, object sizes, maneuver hypotheses, and a time-window protocol.

6 Density and entropy require transported partitions

Let $K_\ell : \mathbb{R}^3 \to [0, \infty)$ satisfy $\int_{\mathbb{R}^3} K_\ell(x)\,dx = 1$, and let $w_i \geq 0$, $\sum_i w_i = 1$. The normalized occupancy density is

$$\rho_\mathfrak{f}(x,t) = \sum_{i=1}^{N} w_i K_\ell(x - y_i(t)). \tag{22}$$

For a radial kernel and a rigid transform $g = (a, R)$, $\rho_g(x,t) = \rho(R^\mathsf{T}(x - a), t)$. This is covariance of a field, not pointwise invariance at one fixed coordinate x.

For a finite measurable partition $\Pi = \{B_1, \ldots, B_M\}$, define

$$p_m = \int_{B_m} \rho(x,t)\, dx, \qquad H_\Pi^{(2)} = -\sum_{p_m>0} p_m \log_2 p_m.$$

The entropy is preserved when Π is transported with the rigid frame. With fixed angular or Cartesian bins it is a protocol-dependent diagnostic. Kernel scale, partition, logarithm base, horizon, and empty-total policy are mandatory fields.

7 Proper time and the synchronization boundary

Use metric signature $(-+++)$. Along a fixed timelike worldline segment γ_i with fixed endpoint events,

$$\tau_i[\gamma_i] = \frac{1}{c}\int_{\gamma_i} \sqrt{-g_{\mu\nu}\, dx^\mu dx^\nu}. \tag{23}$$

This scalar is invariant under coordinate changes. In flat spacetime, the corresponding event interval is invariant under the Poincaré group. Equal-time Euclidean distances are not Lorentz invariants because a boost changes simultaneity.

In a declared stationary weak-field Earth chart with $\rho_i = \|r_i - c_E\|$, $\Phi_i = -\mu_E/\rho_i$, and $v \ll c$,

$$\frac{d\tau_i}{dt} = 1 + \frac{\Phi_i}{c^2} - \frac{\|v_i\|^2}{2c^2} + O(c^{-4}). \tag{24}$$

Relative to an ideal coordinate-stationary reference clock at $R_\star$,

$$\delta_i^{(t,\Phi)} = \frac{d\tau_i}{dt} - \frac{d\tau_\star}{dt} \simeq \frac{\mu_E}{c^2}\left(\frac{1}{R_\star} - \frac{1}{\rho_i}\right) - \frac{\|v_i\|^2}{2c^2}. \tag{25}$$

For a circular two-body orbit,

$$\delta_{\text{circ}}(r) = \frac{\mu_E}{c^2}\left(\frac{1}{R_\star} - \frac{3}{2r}\right). \tag{26}$$

Proposition 7.1 (A spatial locus does not determine proper time)**.** *An unparameterized spatial image $K = \{r(t)\}$ does not determine τ.*

Proof. The same circle can be traversed with two different speed histories. The spatial image is unchanged, while the kinetic term in equation (24) changes. A time-stamped trajectory in a known chart and a declared metric contains strictly more information than its image. □

The comparison $\tau_i(T) - \tau_j(T)$ for endpoints selected by "the same coordinate time T" is protocol-dependent: the time coordinate, slicing, reference potential, and synchronization convention select the endpoint events. This does not contradict scalar invariance of each fixed worldline integral.

8 Benchmark controls and claim firewall

The reproducibility bundle propagates a synthetic labeled network under model 1.1. It checks time-dependent rigid-frame distance invariance, Euclidean-distance-matrix reconstruction, spherical segment clearance, graph invariance under co-transformed geometry, Laplacian spectral invariance under relabeling, exact sampling aliases, temporal earliest arrival, and weak-field clock rates. The numerical values are regression controls, not calibrated constellation predictions.

Table 1: Typed status of the principal legacy quantities.

Object	Transformation or protocol	Correct status
Distance matrix $D(t)$	$C^1(I, \mathrm{SE}(3))$, common time	Invariant when every node is transformed diagonally.
Coordinate trace $y_i(t)$	chosen frame $\mathfrak{f}$	Covariant representation, not invariant.
Spatial closure K_F	constant rigid isometry	Equivariant up to congruence; generally changes for $g(t)$.
Phase closure $\Gamma_{\theta_0,\omega}$	fixed dynamics and time parameter	Model-defined coset subtorus.
Coordinate spectrum	frame plus sampling schedule	Protocol-dependent and aliased.
LOS/link graph	co-transformed body, fixed thresholds	Snapshot invariant; temporal reachability needs ordered contacts.
Proper time $\tau[\gamma]$	fixed worldline endpoints and metric	Spacetime scalar.
Clock difference at $t = T$	declared synchronization and potential	Protocol-dependent comparison.

Boundary 8.1 (Claim boundary). This reference proves elementary transformation, closure, sampling, and graph statements inside declared models. It does not provide operational orbit determination, high-fidelity force models, covariance propagation, collision probability, communication-link budgets, clock steering, or a new theory of satellite networks.

The minimum reproducible protocol records:

(i) labeled state variables, dynamics, constants, force model, epoch, reference frame, orientation convention, and time scale;

(ii) observer origin and attitude, sensor type, light-time policy, occulting-body model, range/elevation masks, and noise model;

(iii) sampling cadence, horizon, missing-data rule, quantizer or partition, estimator, tolerances, and uncertainty statement;

(iv) for clocks, metric signature, potential convention, coordinate time, synchronization, endpoint events, and retained order in c^{-1}.

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